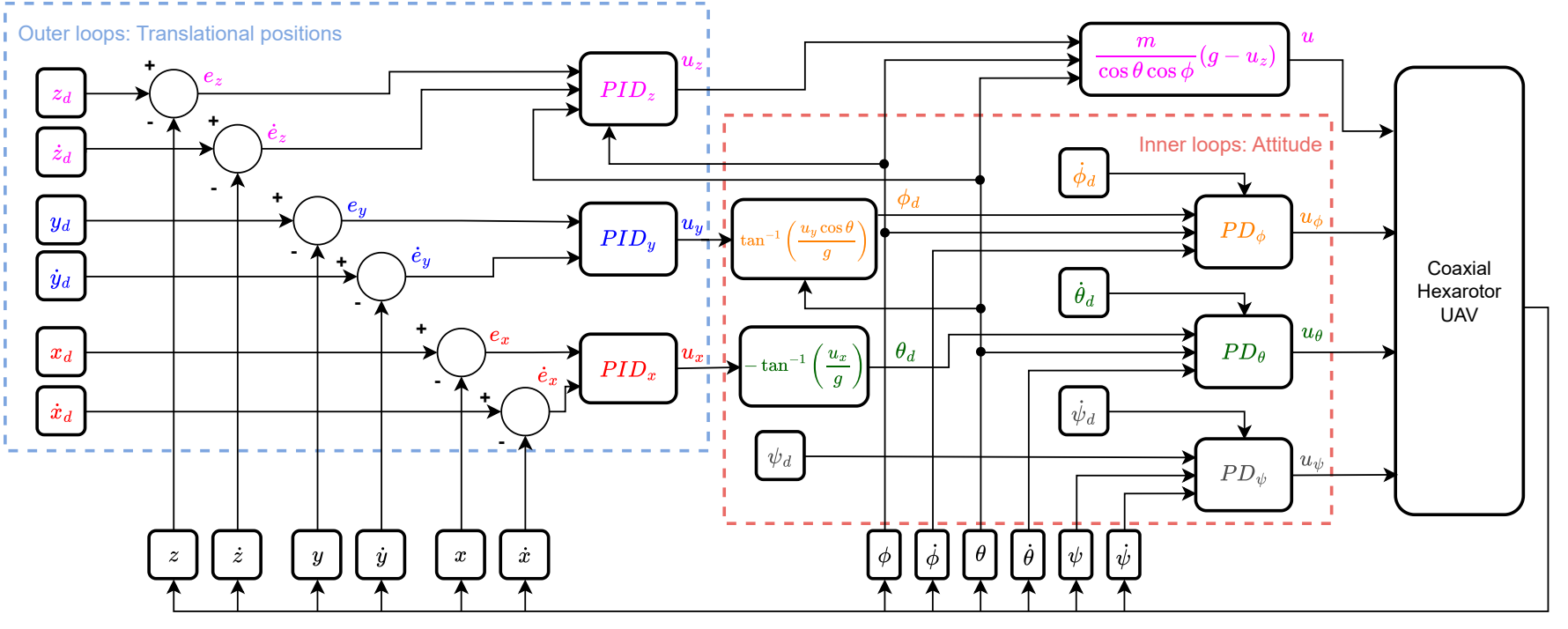

Outer loops: Translational positions
Inner loops: Attitude
$z_d$
$\dot{z}_d$
$y_d$
$\dot{y}_d$
$x_d$
$\dot{x}_d$
$e_z$
$\dot{e}_z$
$e_y$
$\dot{e}_y$
$e_x$
$\dot{e}_x$
$PID_z$
$PID_y$
$PID_x$
$u_z$
$u_y$
$u_x$
$\frac{m}{\cos\theta\cos\phi}(g-u_z)$
$u$
$\tan^{-1}\left(\frac{u_y\cos\theta}{g}\right)$
$-\tan^{-1}\left(\frac{u_x}{g}\right)$
$\phi_d$
$\theta_d$
$\psi_d$
$\dot{\phi}_d$
$\dot{\theta}_d$
$\dot{\psi}_d$
$PD_\phi$
$PD_\theta$
$PD_\psi$
$u_\phi$
$u_\theta$
$u_\psi$
Coaxial Hexarotor UAV
$z$
$\dot{z}$
$y$
$\dot{y}$
$x$
$\dot{x}$
$\phi$
$\dot{\phi}$
$\theta$
$\dot{\theta}$
$\psi$
$\dot{\psi}$